Milestone 04

FirstName LastName

IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

Data Import & Cleaning

Import your dataset and perform essential data cleaning steps.

This includes:

- Recoding and labeling categorical variables
- Handling missing values appropriately
- Identifying and addressing outliers
- Ensuring variables are in the correct format (e.g., factors, numeric)

Document each change to your data clearly so your workflow is fully reproducible.

Table 01

Create a well-formatted and publication-ready Table 01 for your research project using the `tbl_summary()` function in R.

Your table should:

- Have a descriptive title
- Follow standard formatting guidelines and best practices for descriptive statistics tables
- Include a Total column summarizing all observations
- Be grouped by a key categorical variable relevant to your analysis
- Be based on variables that are appropriately cleaned, with missing values handled and outliers addressed prior to summarization

Ensure clarity, consistency, and visual appeal in your table presentation. Document the production of your table (e.g., show a code chunk) so your workflow is fully reproducible.

Data Visual

Use `ggplot2` to visually show the relationship between your dependent variable and 2 predictors:

- One continuous variable
- One categorical variable

You may choose to create a single plot that incorporates both predictors or produce two separate plots, each highlighting one predictor's relationship with the DV.

Your visualizations should:

- Follow best practices in data storytelling and plot design
- Include appropriate labels, scales, and color schemes
- Be clear, interpretable, and tailored to your research context

Aim to communicate insights effectively and support your analytical narrative with compelling visuals.

NOTE: No checkpoints for this milestone.

Simple Regression

- Run a bi-variate regression using your DV and key IV (continuous or categorical).
- Interpret a, b, and R2 from your ouput.

[Show a code chunk demonstrating your regression syntax and output.]